

spring boot microservice notification service

- Spring Boot microservice notification service handles sending various types of notifications (email, SMS, push notifications, in-app notifications) to users in a distributed system.
- This service centralizes notification logic, allowing other microservices to trigger notifications without directly managing the sending mechanisms.
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Key Components and Design Considerations:

- **Event-Driven Architecture:**
 - **Message Broker (e.g., RabbitMQ, Kafka):** Other microservices publish notification events (e.g., "order created," "payment successful") to a message broker.
 - **Notification Service Consumer:** The notification service consumes these events from designated queues or topics.
- **Notification Types:**
 - **Email:** Utilizes Spring Mail Sender for sending emails. Configuration for mail servers (SMTP) is typically done in application.properties.
 - **SMS:** Integrates with third-party SMS gateways (e.g., Twilio, Nexmo) through their respective APIs.
 - **Push Notifications:** Integrates with platforms like Firebase Cloud Messaging (FCM) or Apple Push Notification Service (APNS) for mobile push notifications.
 - **In-app/Real-time Notifications:** Employs WebSockets (e.g., Spring WebSocket with STOMP) to push notifications directly to connected client applications.

Notification Preferences and User Management:

- **User Preferences:** Stores user preferences for notification channels (e.g., email vs. SMS, specific notification types). This can be managed within the notification service or a dedicated user profile service.
- **User Data:** Accesses user information (email address, phone number, device tokens) to target notifications.
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- **Notification Template Management:**

- **Templates:** Uses templates (e.g., Thymeleaf, Freemarker for email) to dynamically generate notification content based on event data.

- **Error Handling and Retries:**

- **Robustness:** Implements error handling and retry mechanisms for failed notification attempts, especially for external integrations (email, SMS, push services).

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Example Flow:

- An "Order Service" processes an order and publishes an "Order Created" event to a RabbitMQ queue.
- The "Notification Service" consumes this "Order Created" event.
- Based on the event data and user preferences, the notification service determines the appropriate notification channels (e.g., email to the customer, SMS to the delivery driver).
- It constructs the notification message using relevant templates and sends it via the respective channels (e.g., JavaMailSender for email, Twilio API for SMS).
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Spring email service

- Spring Boot provides robust **auto-configuration for sending emails** through the spring-boot-starter-mail module and the JavaMailSender interface.
- This functionality simplifies the process of integrating email services into your application for tasks like user notifications, password resets, and confirmations

Key Components

- `spring-boot-starter-mail`: The primary dependency that brings in all necessary JavaMail API libraries and enables Spring Boot's email auto-configuration.
- `JavaMailSender`: The core interface provided by the Spring Framework for sending emails, which can handle simple text messages as well as complex MIME messages with attachments and HTML content.
- `application.properties` (or `.yml`): This file is used to specify the mail server (SMTP) configuration details, which Spring Boot automatically uses to configure the `JavaMailSender` bean.

Implementation Steps

- Add the Dependency

Include the spring-boot-starter-mail dependency in your Maven (pom.xml) or Gradle (build.gradle) file.

Maven (pom.xml):

xml

```
<dependency>
```

```
  <groupId>org.springframework.boot</groupId>
```

```
  <artifactId>spring-boot-starter-mail</artifactId>
```

```
</dependency>
```

Configure Email Properties

- Configure your SMTP server settings in your application.properties file, including host, port, username, and password.
 - For services like Gmail, you may need an "App Password".
- Create an Email Service
 - Create a service class and inject JavaMailSender.
 - You can define methods to send simple emails using SimpleMailMessage or more complex emails with HTML or attachments using MimeMessageHelper.

application.properties

- In your application.properties or application.yml, configure the SMTP server details.
- Replace the placeholders with your actual email server information:

spring.mail.host=smtp.example.com

spring.mail.port=587 # or 465 for SSL/TLS

spring.mail.username=your-email@example.com

spring.mail.password=your-email-password

spring.mail.properties.mail.smtp.auth=true

spring.mail.properties.mail.smtp.starttls.enable=true # or
ssl.enable=true for SSL

Create an Email Service:

@Service

```
public class EmailService {
```

@Autowired

```
private JavaMailSender javaMailSender;
```

```
    public void sendSimpleEmail(String to, String subject, String  
body) throws MessagingException {
```

```
        MimeMessage mimeTypeMessage =  
javaMailSender.createMimeMessage();
```

```
        MimeMessageHelper helper = new  
MimeMessageHelper(mimeTypeMessage, "UTF-8");
```

```
        helper.setFrom("your-email@example.com"); // Set your  
sender email
```

```
        helper.setTo(to);
```

```
        helper.setSubject(subject);
```

```
        helper.setText(body);
```

```
        javaMailSender.send(mimeTypeMessage);
```

```
        System.out.println("Email sent successfully to " + to);
```

```
}
```

```
    public void sendEmailWithAttachment(String to, String subject,  
String body, String attachmentPath, String attachmentName)  
throws MessagingException {
```

```
        MimeMessage mimeTypeMessage =  
javaMailSender.createMimeMessage();
```

```
        MimeMessageHelper helper = new  
MimeMessageHelper(mimeTypeMessage, true, "UTF-8"); // true for  
multipart
```

```
        helper.setFrom("your-email@example.com");
```

```
        helper.setTo(to);
```

```
        helper.setSubject(subject);
```

```
        helper.setText(body, true); // true for HTML content
```

```
        // Add attachment
```

```
        // FileSystemResource file = new FileSystemResource(new  
File(attachmentPath));
```

```
        // helper.addAttachment(attachmentName, file);
```

```
        javaMailSender.send(mimeTypeMessage);
```

```
        System.out.println("Email with attachment sent successfully  
to " + to);
```

```
}
```

```
}
```

Use the Email Service:

- Inject the EmailService into your controllers or other service classes to send emails:

@RestController

```
public class EmailController {
```

```
    @Autowired
```

```
    private EmailService emailService;
```

```
    @GetMapping("/send-email")
```

```
    public String sendEmail() {
```

```
        try {
```

```
            emailService.sendSimpleEmail("recipient@example.com", "Test  
Subject", "Hello from Spring Boot!");
```

```
            return "Email sent!";
```

```
        } catch (MessagingException e) {
```

```
            e.printStackTrace();
```

```
            return "Error sending email: " + e.getMessage();
```

```
        }
```

```
    }
```

```
}
```